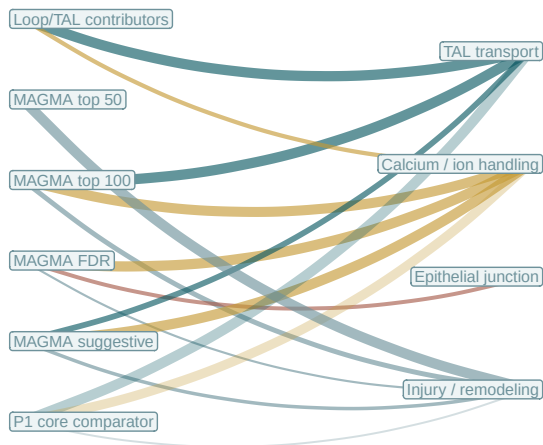


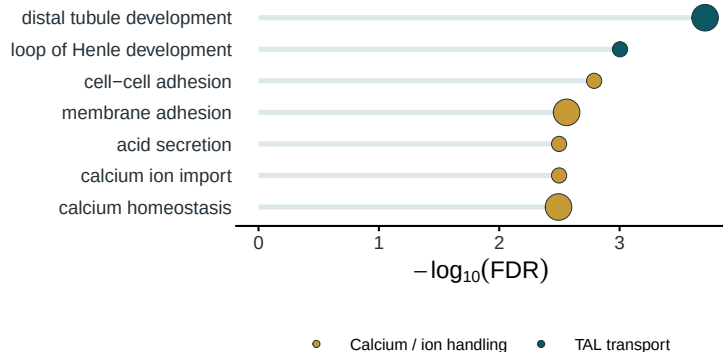
A. Functional-context evidence network

Link width is relative within each evidence layer; not a causal effect scale



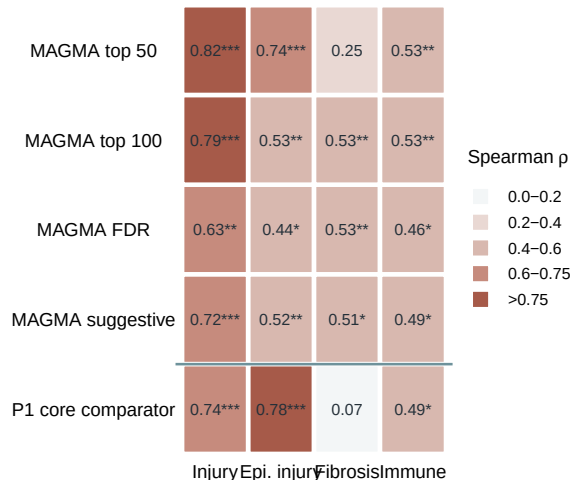
B. Ranked non-redundant GO context

Seven representative terms; point size = gene count



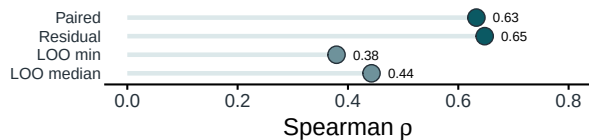
C. Risk-injury module coupling

Paired-delta Spearman rho
Stars indicate FDR thresholds



D. Robustness and extension audit

Coupling checks above; resource audit below



TWAS	SMR/coloc	Spatial
scaffold ready	scaffold ready	scaffold ready
resource missing	resource missing	resource missing
not used for claims	not used for claims	not used for claims